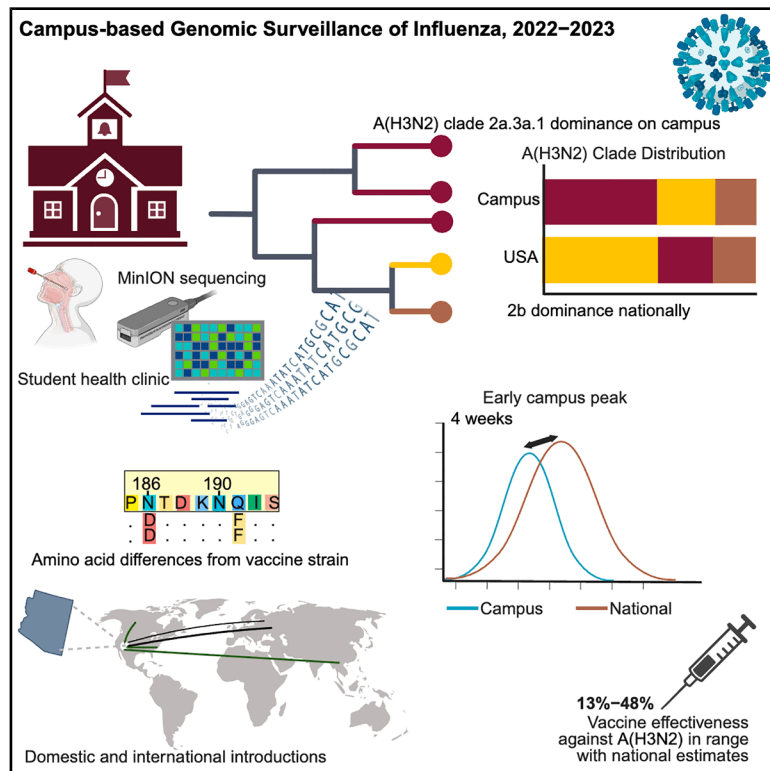


Campus-based genomic surveillance uncovers early emergence of a future dominant A(H3N2) influenza clade

Graphical abstract



Authors

Matthew Scotch, Temitope O.C. Faleye, Jillian M. Wright, Sarah Finnerty, Rolf U. Halden, Arvind Varsani

Correspondence

matthew.scotch@asu.edu

In brief

Immunology; Virology

Highlights

- Genomic surveillance shows influenza A(H3N2) 2a.3a.1 campus dominance for 2022–2023
- Campus A(H3N2) clade diversity contrasts with national dominance of A(H3N2) 2b
- Clade introductions from North America and Europe supported local transmission
- Epitope-based VE 0.13–0.48 aligns with national estimates for the 2022–2023 season



Article

Campus-based genomic surveillance uncovers early emergence of a future dominant A(H3N2) influenza clade

Matthew Scotch,^{1,2,5,*} Temitope O.C. Faleye,¹ Jillian M. Wright,¹ Sarah Finnerty,³ Rolf U. Halden,¹ and Arvind Varsani⁴

¹Biodesign Center for Environmental Health Engineering, Arizona State University, Tempe, AZ, USA

²College of Health Solutions, Arizona State University, Phoenix, AZ, USA

³Health Services, Arizona State University, Tempe, AZ, USA

⁴Biodesign Center for Fundamental and Applied Microbiomics, Center for Evolution and Medicine and School of Life Sciences, Arizona State University, Tempe, AZ 85287, USA

⁵Lead contact

*Correspondence: matthew.scotch@asu.edu

<https://doi.org/10.1016/j.isci.2025.113941>

SUMMARY

We conducted genomic surveillance during the 2022–2023 northern hemisphere influenza season at a university in the southwestern US to assess how local diversity and evolution compared to national trends. We identified 100 positive samples (19%) from clinical swabs collected at a student health clinic and observed a dominance of subtype A(H3N2) which was consistent nationally for that season. However, we found stark differences when examining H3 clades, including an early dominance of 3C.2a1b.2a.2a.3a.1 variants contrasting from country level data in which 3C.2a1b.2a.2b variants were most abundant. We found known H3 amino acid substitutions in epitope sites and estimated vaccine effectiveness to be 0.13–0.48 which aligns with national estimates for that season. Importantly, 3C.2a1b.2a.2a.3a.1 became the dominant H3 clade nationally in 2023–2024 and 2024–2025, highlighting the potential for localized genomic surveillance in semi-closed, highly dense university settings and its potential for early insight of seasonal influenza diversity.

INTRODUCTION

Genomic epidemiology has transformed influenza surveillance by enabling real-time tracking of viral evolution, diversity, and spread.^{1,2} This approach informs vaccine strain selection, antiviral resistance monitoring, and outbreak response. University campuses are ideal for such studies due to their high population density, centralized health services, and social connectivity, which facilitate transmission and structured sampling.

Influenza A viruses (IAVs), responsible for both seasonal epidemics and pandemics, are classified by the antigenic properties of their surface glycoproteins, hemagglutinin (HA), and neuraminidase (NA), which are major targets of host immunity. These proteins evolve rapidly through antigenic drift, complicating vaccine effectiveness and necessitating continual surveillance.^{3,4} The exceptional genetic diversity, rapid rate of evolution, and zoonotic potential of IAVs make them a persistent public health challenge.^{5–7}

Although influenza activity has rebounded following the COVID-19 pandemic, only a few post-pandemic, genomic epidemiology studies have focused on university settings for understanding evolution and spread. Past efforts have included students in broader populations^{8,9} or focused on shorter time frames,¹⁰ but there remains a gap in understanding how influenza evolves in densely populated campus environments. To address this, we conducted genomic epidemiology of influenza

viruses at the Arizona State University (ASU) during the 2022–2023 season, which featured an early surge in cases and a rapid post-holiday decline. With an enrollment of over 78,000 students on campus,¹¹ ASU offers a unique opportunity to assess intra-campus transmission, viral diversity, and evolutionary dynamics in a semi-closed population. By comparing these findings with national patterns, we aim to inform surveillance strategies and public health preparedness for other high-risk, congregate settings.

RESULTS

In [Figure 1](#), we show the final determination of the 516 nasal swab samples that we analyzed from the university's student health clinic on the Tempe campus during the 2022–2023 season.

For the 100 positives (19%), we identified 91 A(H3N2), 7 A(H1N1)pdm09, and 2 B(Victoria). Meanwhile, 416 samples tested negative. We note that while H3 was confirmed in 91 samples, N2 was only confirmed in 76 of these; therefore, it is possible (though unlikely) that not all the 15 samples lacking an N2 sequences were A(H3N2). In [Figure S1](#), we show the results by week of collection starting from week 43 in 2022 to week 19 in 2023. In November, the early peak in cases on campus occurred four weeks before (44 compared to 48) the national peak ([Figure S2](#)). The earlier season peak was likely a contributing



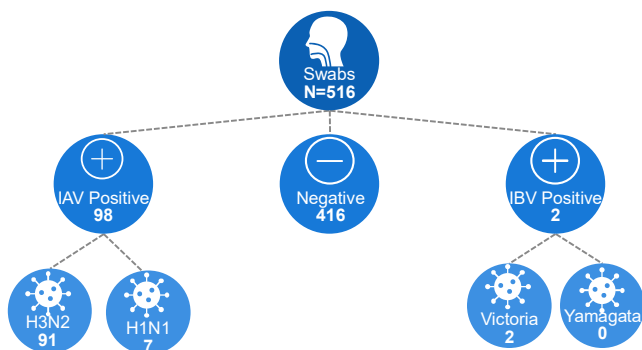


Figure 1. Results of our 516 clinical swabs collected at the student health clinic

Providers tested samples at the student health clinic as part of routine clinical care and labeled collection tubes as A, B, A/B, or negative, based on results from a lateral flow immunoassay. We then subtyped positives via real-time PCR and pathogen sequencing. Abbreviations: IAV = influenza A virus; IBV = influenza B virus. Note: while H3 was confirmed in 91 samples, N2 was only confirmed in 76 of these; therefore, it is possible that not all H3 samples were A(H3N2).

factor to the sharp decline in cases after 2023 which resulted in almost all negative samples on campus.

For our 100 positive university samples, we generated 2,042,446 raw reads, of which 1,710,494 (83.75%) passed quality thresholds, and 1,525,574 (89.19%) of those were successfully mapped by IRMA. The median coverage depth varied across gene segments, with H3 exhibiting by far the highest mean depth (Figure S3). All 91 H3 segments met the coverage threshold of 100, while 76 out of 87 (87%) N2 segments satisfied the criteria (Table S1). Through variant analysis, we identified synonymous and missense mutations for H3 (Figure S4), N2 (Figure S5), H1 (Figure S6), and N1 (Figure S7). For H3, we identified 2,182 variants, the majority of which were synonymous (1,290 or 59%), followed by missense mutations (892 or 41%). N2 showed a similar distribution among its 1,086 variants, with most being synonymous (802 or 74%) in comparison to missense (284 or 26%).

High sequence similarity and divergent clade patterns

Throughout this manuscript, we refer to clades of the HA (H3) segment using the standard “3c’ nomenclature” (e.g., 3C.2a1b.2a), consistent with common practice. In contrast, for the NA (N2) segment, we follow Nextstrain’s nomenclature¹² for subclades that designate major lineages and sub-lineages (e.g., clades A and B, subclades A.2.2, B.3, etc.).

We show strong sequence similarity for our H3 (Figure S8) and N2 (Figure S9) sequences, with average identity scores of 0.990 and 0.989, respectively. Fifty-six of the 91 university H3 sequences (62%) belonged to clade 3C.2a1b.2a.2a.3a.1, 20 (22%) to 3C.2a1b.2a.2b, and 11 (12%) to 3C.2a1b.2a.2a.1 (Figure S10A). The remaining four sequences (4%) represented 3C.2a1b.2a.2a, 3C.2a1b.2a.2a.1b, 3C.2a1b.2a.2a.3, and 3C.2a1b.2a.2a.3b. Notably, while clade 3C.2a1b.2a.2b was the dominant lineage nationally during the same season (accounting for over 70% of sequences at times),¹³ it was a minority on campus, suggesting localized transmission dynamics that

diverged from national trends. For N2 (Figure S10B), 63% of sequences (48/76) were classified as B.4, followed by 20% (15/76) as A.2.2.2, with the remainder representing B.2 (6), B.2.2 (5), and B (1).

In contrast to the dominance of A(H3N2) at both the university and national levels, A(H1N1)pdm09 was far less prevalent in our campus dataset. While A(H3N2) comprised 93% of our A/HA sequences, only 7% were H1 and in comparison to 27% nationally during the 2022–2023 season.¹⁴ Like H3 and N2, our H1 and N1 sequences exhibited strong similarity (Figure S11) with an average identity score of 0.988 and 0.993, respectively. Four of the seven university H1 sequences (57%) belonged to 6B.1A.5a.2a.1, while three of the sequences (43%) were 6B.1A.5a.2a (Figure S12A). For N1 (Figure S12B), three of the seven were C.5.1.1 (43%), two were C.5.3 (29%), and there was one of C.2 (14%) and C.5.1 (14%) each.

In addition to influenza A, we also identified two influenza B/Victoria variants (2% of our samples) that were both representative of clade V1A.3a.2, which has dominated the B/Victoria landscape for several years. The sequences belonged to different subclades, however, with one in C.3 and one in C.5.7.

For our Bayesian phylogenetic inference and epidemiologic modeling, we focused on subtype A(H3N2), which accounted for a significant proportion of the influenza variants circulating on campus during the study period. Its high prevalence, combined with the observed clade diversity, provided a robust foundation for downstream analyses of evolutionary trajectories and epidemiological dynamics.

Distinct introductions drive shifts in dominant clades

We integrated our university-derived sequences with representative data from the Global Initiative on Sharing All Influenza Data (GISAID) database,¹³ resulting in a comprehensive dataset of 395 H3 and 377 N2 sequences. In Figure 2A, we present the sub-tree of the full phylogeny (Figure S13) that includes all sequences descended from the most recent common ancestor (TMRCA) of our campus samples. For H3, we estimated the median TMRCA to be 3.22 years (95% highest posterior density [HPD], 3.00–3.47 years), corresponding to approximately February 28, 2020 (95% HPD, 27 November 2019–18 May 2020). This predates the 2020–2021 northern hemisphere influenza season and during the pandemic period, despite historically low influenza circulation. We estimated the median rate of evolution across all H3 clades at 3.23×10^{-3} substitutions/site/year (95% HPD, 2.94×10^{-3} – 3.55×10^{-3}).

Of the 215 sequences in the sub-tree (Figure 2A), 91 originated from the university, with representation across eleven distinct H3 clades. Six of these clades (3C.2a1b.2a.2a.1a, 3C.2a1b.2a.2a.1b, 3C.2a1b.2a.2a.2, 3C.2a1b.2a.2a.3a.1, 3C.2a1b.2a.2a.3b, and 3C.2a1b.2a.2b) were monophyletic, consistent with discrete introduction events. Clade 3C.2a1b.2a.2a.3a.1 was the most common among early-season variants, while clade 3C.2a1b.2a.2b emerged prominently later, forming a large, monophyletic cluster suggesting the occurrence of extensive local transmission on campus.

In panel B in Figure 2, we show the timing, origin, and clades for inferred introduction events. We define an introduction event as a branch in the maximum clade credibility tree where a

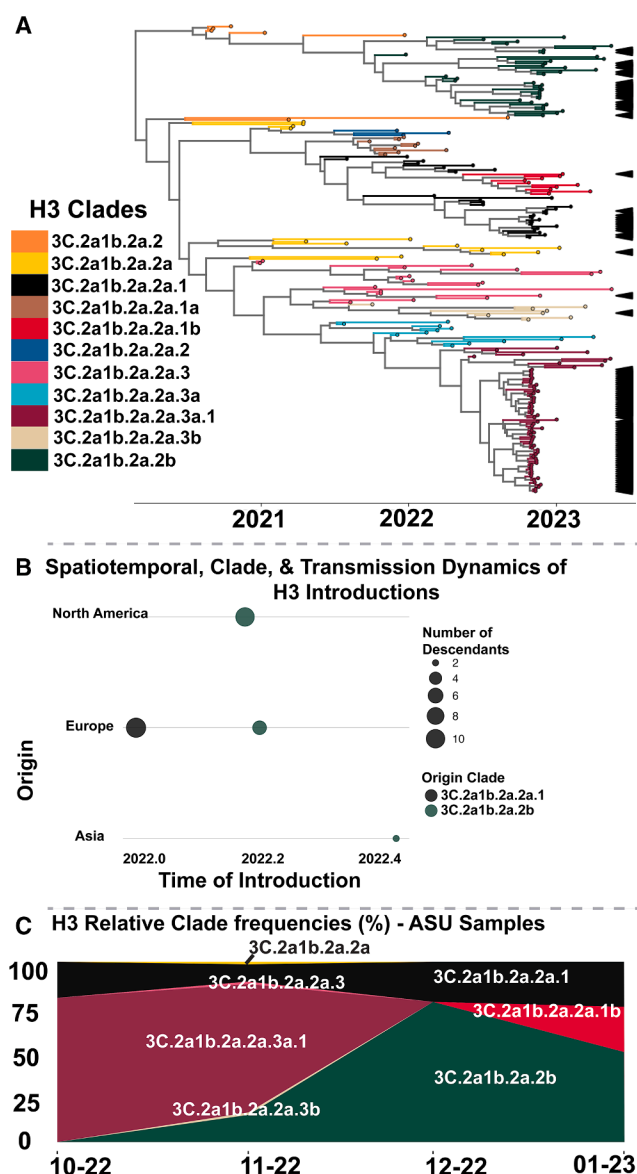


Figure 2. Spatiotemporal and evolutionary dynamics of H3 influenza clades circulating on campus

(A) We show a sub-tree of the MCC tree from Figure S13, pruned to include only sequences descended from the TMRCA of the ASU samples. Black arrows indicate ASU sequences ($n = 91$); all other tips represent GISAID sequences.

(B) We summarize the timing, geographic origin, and transmission dynamics of H3 introductions into campus, with circle sizes reflecting the number of sampled descendants per introduction.

(C) We display the monthly distribution of H3 clade frequencies among our university samples from October 2022 to January 2023 using the color scheme depicted in (A). Axis labels and legends finalized in Adobe Illustrator.

Markov jump into the university was inferred from a non-university parent lineage. Here, each point corresponds to a unique introduction, sized by the number of downstream descendants and colored by clade. The four distinct events highlight introduc-

tions from Asia, Europe, and North America. Introduction events linked to clade 3C.2a1b.2a.2a.1 from Europe resulted in the largest number of descendants ($n = 11$), while a North America-derived 3C.2a1b.2a.2b introduction yielded the second largest amount of observed local transmission ($n = 10$).

We see this replacement pattern in Figure 2C, which tracks the changing distribution of clades over time. Clade 3C.2a1b.2a.2a.3a.1 (maroon color) dominated in October and November 2022 but was quickly overtaken by clade 3C.2a1b.2a.2b (green) in December. The 3C.2a1b.2a.2b clade maintained its dominance until influenza activity declined sharply in January 2023 (Figure S1), marking the end of the season on campus.

In Figure 3A, we show the sub-tree of the full N2 maximum clade credibility (MCC) phylogeny (Figure S14) representing all sequences descended from TMRCA of the university samples. The median TMRCA for N2 was estimated to be 6.19 years (95% HPD, 5.71–6.75 years), corresponding to 9 March 2017 (95% HPD, 18 August 2016–2 September 2017). The estimated median evolutionary rate was 2.96×10^{-3} substitutions/site/year (95% HPD, 2.63×10^{-3} – 3.31×10^{-3}).

The MCC tree reveals two major lineages of N2, further subdivided into distinct subclades. Of the 298 sequences in the sub-tree, 76 were derived from the university and distributed across ten unique subclades. Several subclades, such as A.2.2.2, B.2.2, B.3, and B.4, formed monophyletic clusters. In contrast, sub-clade A.2.2 appeared in multiple branches of the tree, consistent with a pattern previously reported in the literature.¹⁵

We identified six distinct N2 introduction events in 2022 (Figure 3B), originating from Asia (B.4), North America (two separate introductions of A.2.2.2 and one of B.2), Europe (B.2.2), and Africa (A.2.2.2). The introduction of B.4 from Asia was the most impactful, generating a large transmission chain with 46 descendant sequences (this is also reflected in Figure 3C), where B.4 dominated circulation in October and November. Two independent introductions of A.2.2.2 from North America produced 11 and 5 sequences, respectively, while the African A.2.2.2 lineage contributed two. The B.2 and B.2.2 introductions yielded 7 and 5 descendants, respectively.

In Figure 3C, we show the proportion of subclades on campus over time. Subclade B.4 (maroon color), which dominated the early season, declined sharply by December. This decline corresponded to the rise of A and then B subclades, A.2.2.2 (black) and B.2.2 (light blue), respectively. In particular, A.2.2.2 rapidly displaced B.4 in November while B.2.2 replaced A.2.2.2 and became the dominant lineages through January 2023. The rise of A.2.2.2 is primarily attributed to two separate introduction events from North America in early 2022, each leading to local transmission. A third introduction occurred at the start of the influenza season but had a much smaller transmission impact. Meanwhile, the rise of B.2.2 appears to be driven in part by a single introduction from Europe in late August, shortly before the start of the season, followed by a limited expansion.

Epidemic peak on campus coincides with clade 3C.2a1b.2a.2b emergence

Using a two-deme PhyDyn SIR model, we estimated a basic reproduction number (R_0) of 1.17 (95% HPD, 1.06–1.30). This lies within, though toward the low end of, reported estimates

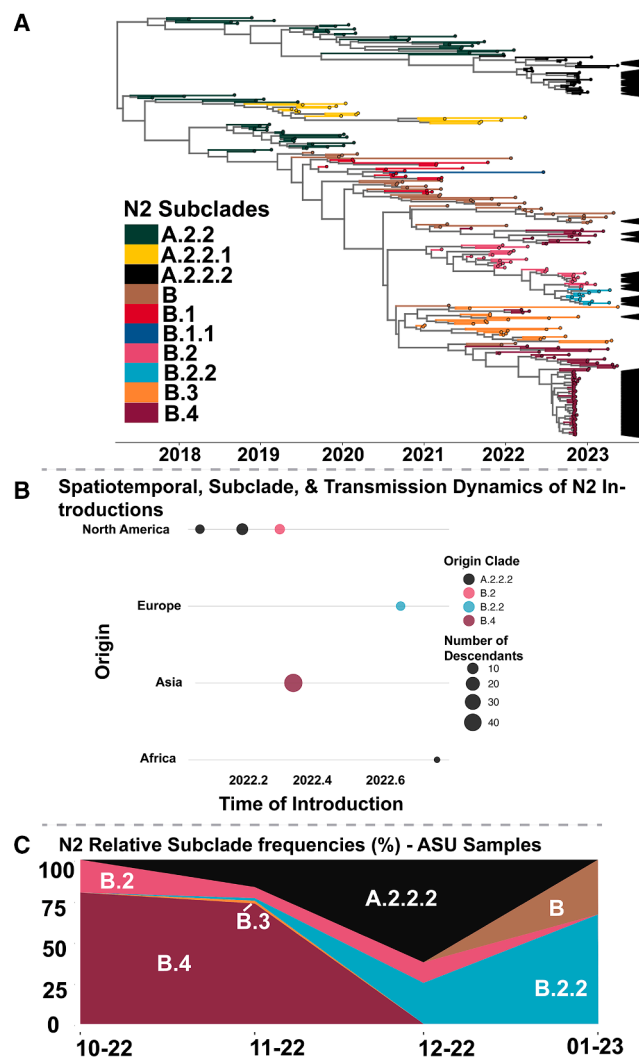


Figure 3. Spatiotemporal and evolutionary dynamics of N2 influenza clades circulating on campus

(A) We show a sub-tree of the MCC tree from Figure S14, pruned to include only sequences descended from the TMRCA of the ASU samples. Black arrows indicate ASU sequences ($n = 76$); all other tips represent GISAID sequences. We color-code branches by subclade using abbreviated names (e.g., A.2.2.1).

(B) We summarize the timing, geographic origin, and transmission dynamics of H3 introductions into campus, with circle sizes reflecting the number of sampled descendants per introduction.

(C) We display the monthly distribution of N2 clade frequencies among our university samples from October 2022 to January 2023 using the color scheme depicted in (A). Axis labels and legends finalized in Adobe Illustrator.

for seasonal influenza.^{16,17} As we show in Figure S15 panel A, the number of susceptible individuals decreased steadily over the semester as students became infected. At the start of the season in early October, the model estimated 123 infected and 132 recovered individuals. The infected population increased over the following weeks, peaking at 581 individuals in late November or early December. The campus peak coincided with the emer-

Table 1. H3 amino acid substitutions with a prevalence of ≥ 0.2

AA Substitution	Count	Proportion ($n = 91$)	Associated clades
G225D	91	1.00	3C.2a1b.2a.2a 3C.2a1b.2a.2a.1 3C.2a1b.2a.2a.3 3C.2a1b.2a.2a.3a.1 3C.2a1b.2a.2a.3b 3C.2a1b.2a.2b
N186D	91	1.00	3C.2a1b.2a.2a 3C.2a1b.2a.2a.1 3C.2a1b.2a.2a.3 3C.2a1b.2a.2a.3a.1 3C.2a1b.2a.2a.3b 3C.2a1b.2a.2b
I140K	77	0.85	3C.2a1b.2a.2a.1 3C.2a1b.2a.2a.3a.1 3C.2a1b.2a.2b
E50K	76	0.84	3C.2a1b.2a.2a.3a.1 3C.2a1b.2a.2b
D53N	59	0.65	3C.2a1b.2a.2a 3C.2a1b.2a.2a.3 3C.2a1b.2a.2a.3a.1 3C.2a1b.2a.2a.3b
I192F	58	0.64	3C.2a1b.2a.2a.3 3C.2a1b.2a.2a.3a.1 3C.2a1b.2a.2a.3b
N378S	58	0.64	3C.2a1b.2a.2a.3 3C.2a1b.2a.2a.3a.1 3C.2a1b.2a.2a.3b
N96S	57	0.63	3C.2a1b.2a.2a.3 3C.2a1b.2a.2a.3a.1
I223V	56	0.62	3C.2a1b.2a.2a.3a.1
V204I	23	0.25	3C.2a1b.2a.2a.3a.1
S156H	20	0.22	3C.2a1b.2a.2b
F79V	18	0.20	3C.2a1b.2a.2b

For each substitution (H3 numbering), we show its count, proportion across our 91 variants, and associated clades. We bold the substitutions in known epitope regions.

gence and rapid rise of clade 3C.2a1b.2a.2b (Figure 2C), although we cannot infer causality.

The dynamics of the epidemic are further reflected in Figure S15B, which shows the rate of change in each compartment. Infection rates had already begun to decline before the end of the fall semester (first dashed line), and we observed no resurgence following the start of the spring semester in January (second dashed line). These patterns suggest that the epidemic largely subsided by early 2023, consistent with national trends (Figure S2) and with a reduced susceptible population remaining on campus over winter break.

Prevalent and clade-specific amino acid substitutions

For our consensus sequences, we identified 777 total amino acid substitutions across our 91 H3 proteins with A/Darwin/9 as our reference strain. This equated to an average of 8.5 substitutions per protein. In Table 1, we show the most prevalent substitutions

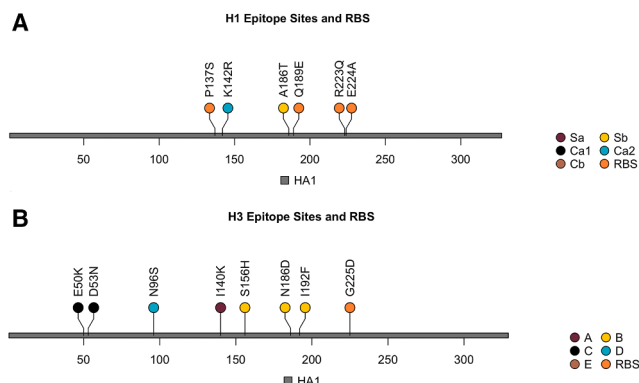


Figure 4. Amino acid substitutions from our university influenza A virus sequences with a prevalence of ≥ 0.2 and are in known epitope and receptor-binding sites

(A) Lollipop plot showing amino acid substitutions for our H1 virus sequences (H1 numbering).

(B) Amino acid substitutions for our H3 virus sequences (H3 numbering). Axis labels and legends finalized in Adobe Illustrator. Abbreviations: RBS = receptor-binding site.

(≥ 0.2) and their 3C.2a1b associated clades. G225 and N186D were found in all H3 variants sampled on campus, while I140K and E50K were present in over 80% (Figure 4B). We also observed eleven substitutions that have been observed in 3C.2a1b.2a.2b including S156H, F79V, D101E, T135A, F79I, A476T, N122S, Q356L, S137A, S54N, and S91N.

We observed 238 total N2 amino acid substitutions from A/Darwin/9 across our 76 proteins for an average of 3.13 variants per protein (Table 2). This includes one instance of substitution S331R which has been showed to result in reduced susceptibility to oseltamivir and zanamivir.¹⁸

For our seven A(H1N1)pdm09 sequences, we identified 97 total H1 (Table S2; Figure 4A) and 20 total N1 (Table S3) amino acid substitutions with A/Victoria/2570/2019 as the reference strain. This equated to an average of 13.9 (H1) and 2.86 (N1) substitutions per protein, respectively. None of the N1 substitutions have been showed to impact antiviral sensitivity (Table S3).

For our two B/Victoria variants, one HA segment had three substitutions including E121G, E169K, and D184E, while the other had four substitutions including E121K, A146E, E185G, and S195P (against reference strain B/Austria/1359417/2021). Regarding NA substitutions, one virus sequence had A44V, G372E, and I459V, while the other had R57H, V388I, and I459V. None of the NA substitutions have been linked to antiviral drug resistance.

Variation in predicted vaccine effectiveness and N-glycosylation sites

Among 25 unique translated H3 sequences from the university, we observed a variation of vaccine effectiveness estimates (0.13–0.48) and dominant epitope regions (Table S4). Region B was the most common dominant epitope impacted (16/25 or 64%), followed by A (5/25, or 20%) and then C (4/25 or 16%). No sequence included D or E as the dominant epitope. We found I140K to be the predominant epitope A substitution (19/25 unique sequences, 76%). Nationally, 87% of the 2022–

Table 2. N2 amino acid substitutions with a prevalence of ≥ 0.2

AA Substitution	Count	Proportion ($n = 76$)
S150H	47	0.62
S150R	29	0.38
V263I	16	0.21
D346S	15	0.20
N463D	15	0.20
R315S	15	0.20
S465N	15	0.20

For each substitution, we show its count and proportion across our 76 variants.

2023 US H3N2 HA sequences in GISAID (4,254/4,884) also carried I140K. For epitope B, we observed N186D in all 25 sequences (100%), consistent with its near-universal national prevalence (4,882/4,884, 99.96%). We identified E50K as the predominant epitope C substitution (18/25, 72%) and detected it in 76.8% of the 2022–2023 US H3N2 HA sequences in GISAID (3,753/4,884). We performed a one-way ANOVA and found marginal differences in predicted effectiveness of influenza vaccines (VE) across the dominant epitope groups, but the differences were not statistically significant ($p = 0.071$, Table S5).

We leveraged NetNGlyc v1.0 to identify asparagine-Xaa-serine/threonine sequons and predict N-glycosylation sites within our 91 H3 sequences. In total, we observed 14 unique sequons (for 13 sites). Nine of the sequons were observed in all sequences, while two were observed in all but one sequence. Four sequons had an average potential >0.7 including NSTA (position 24), NGTI (38), NCTL (79), and NVTM (181). In Table 3, we show the sequons with unanimous jury agreement across nine neural networks (ordered by position on the H3 protein). Of note, NTSA (position 24), NCTL (position 79), and NGSI (position 301) were predicted as N-glycosylated and found in all sequences with high specificity. Not surprisingly, all sequons listed in Table 3 are found in the HA1 subunit (globular head).

DISCUSSION

Universities represent an important, yet understudied environment for influenza genomic epidemiology due to their semi-closed environment, close mixing of individuals, and potential early indication of state and national-level trends. The 2022–2023 influenza season in the United States signified a near-return to pre-pandemic levels¹⁰ as well as a historically early peak in clinical cases resulting in an early population bottleneck and end to the influenza season. The confluence of these factors makes the 2022–2023 season an important time-point in university-focused influenza evolution and dynamics.

We conducted genomic surveillance of seasonal influenza viruses during the 2022–2023 northern hemisphere season on a large university setting in the southwestern United States to understand the diversity, evolution, and spread within a local environment and in comparison to national and international trends. We focused on the amplification and sequencing of HA and NA gene segments for influenza A and B seasonal viruses. We

Table 3. Summary of H3 sequons with glycosylation potential among our 91 H3 samples as estimated by NetNGlyc - 1.0

Position	Sequon	Count	Mean Potential	N-Glyc Result
24	NSTA	91	0.79	+++
38	NGTI	90	0.70	++
38	NGTM	1	0.66	++
79	NCTL	91	0.74	++
149	NGTS	14	0.64	++
1881	NVTM	90	0.75	++
262	NSTG	3	0.64	++
301	NGSI	91	0.68	++

We include sequons with high specificity and unanimous agreement across all nine neural networks and ordered by position on the HA protein (H3 numbering). N-Glyc Result key: '++++' = Potential ≥ 0.90 and Jury agreement; '+++ = Potential ≥ 0.75 and Jury agreement; '++' = Potential ≥ 0.5 and Jury agreement or Potential ≥ 0.75 ; '+' = Potential ≥ 0.5 .¹⁹

observed a dominance of influenza A(H3N2) (Figures 1 and S1), which was consistent with national trends for that season.¹⁴ However, we found stark differences when examining subtype-specific H3 clades. On campus, 3C.2a1b.2a.2a.3a.1 predominated early in the season (Figure 2), in contrast to national-level data where 3C.2a1b.2a.2b was most abundant. These variants may have contributed to the early campus peak relative to national trends (Figure 2), although causality cannot be inferred. Notably, a similar pattern was observed in Europe in 2016–2017, where the season started three weeks early and peaked in late December–early January while A/H3N2 predominated. Among genetically characterized A/H3N2 viruses, 71% belonged to the emergent 3C.2a1 clade.²⁰ As 3C.2a1b.2a.2a.3a.1 variants became nearly the exclusive H3 clade nationally in 2023–2024²¹ as well as 2024–2025,²² our identification of their presence on campus underscores the value of local genomic surveillance as a potential early indicator of broader influenza trends.

We used phylodynamics to understand the timing, source, and impact of clade-specific introductions on campus (Figure 2). Interestingly, we did not observe any 3C.2a1b.2a.2a.3a.1 introductions suggesting the potential for earlier cryptic transmission²³ or as a result of insufficient sampling potentially exacerbated by the early peak on campus. We did, however, find introductions of 3C.2a1b.2a.2b variants from North America, Europe, and Asia in early 2022 which possibly contributed to its later-season local dominance toward the end of 2022. The rise of clade 3C.2a1b.2a.2b was also impactful in our Bayesian epidemiological model, as we observed a peak in the population of infected individuals on campus during its emergence and rapid rise (Figure S15).

As H3 sequences reveal an important, yet incomplete, explanation of seasonal influenza evolution and dynamics, we also studied N2 subclades (Figures 3 and S10) and found a similar pattern of early dominance of one group followed by a replacement by another, in B.4 and A.2.2.2 respectively. Both subclades formed monophyletic clusters on our MCC tree (Figure 3) potentially due to introduction events, albeit from different sources. The early dominance of B.4 variants was likely influenced by a

single introduction event in early 2022 from Asia which led to significant downstream transmission. Meanwhile, we observed two introduction events for the A.2.2 subclade including one from Africa right around the start of the influenza season which potentially contributed to its eventual takeover of B.4 variants on campus.

We utilized Bayesian phylogenetic inference to estimate the TMRCA of our university H3 and N2 sequences and estimated the median TMRCA of H3 to be 3.22 years (95% HPD, 3.00–3.47 years), which predates the 2020–2021 northern hemisphere influenza season. Correspondingly, we estimated the median rate of evolution across all H3 clades at 3.23×10^{-3} substitutions/site/year (95% HPD, 2.94×10^{-3} – 3.55×10^{-3}), consistent with some previous reports,^{24,25} but slower than others.^{26,27} For N2, we estimated a median evolutionary rate of 2.96×10^{-3} substitutions/site/year (95% HPD, 2.63×10^{-3} – 3.31×10^{-3}), consistent with previous reports,^{26,28,29} while the median TMRCA was 6.19 years (95% HPD, 5.71–6.75 years), corresponding to 9 March 2017 (95% HPD: 18 August 2016–2 September 2017). Overall, the slower estimated rates of evolution might be partially explained by the time-dependent rate phenomenon³⁰ in which the presence of historical samples over longer timescales may reduce the rate estimate of more recent samples. We observed the opposite after running a separate Bayesian inference model that was restricted to our 91 university H3 sequences from the 2022–2023 season and recorded a higher evolutionary rate of 4.06×10^{-3} substitutions/site/year (95% HPD, 3.14×10^{-3} – 5.09×10^{-3}). In addition, we note that the nearly three-year difference between the estimated TMRCA of HA and NA, with the former younger than the latter, may be explained by gene segment reassortment and HA-specific selective sweeps.^{31,32}

We found several highly prevalent H3 substitutions that were noted in other surveillance efforts during the 2022–2023 season^{33,34} including N186 which was present in all of our 91 H3 variants (Table 1) and is also in a known epitope region (Figure 4B). In addition, we identified substitutions G225, N186, and I140K in most of our H3 sequences, which have been shown to make the surface of the receptor binding domain more negatively charged than the vaccine strain.³⁵ We also note the high prevalence (63%) of N96S which is a defining amino acid substitution of 3C.2a1b.2a.2a.3 and 3C.2a1b.2a.2a.3a.1 variants and creates a new potential N-glycosylation site in the globular head.³⁶ For N2, we found one instance of S331R (Figure S5) which has been showed to result in reduced susceptibility to oseltamivir and zanamivir.¹⁸

We estimated vaccine effectiveness using an H3 epitope model and observed a range of 0.13–0.48 which overlaps with published estimates for that season.^{37,38} Multiple substitutions at H3 antigenic sites, together with several putative N-linked glycosylation sequons in HA1 (globular head; Table 3), could plausibly contribute to a moderate VE, although predicted sequons do not guarantee the presence of glycans. In addition, since we lack student vaccination data for 2022–2023, we are unable to determine whether, or by how much, vaccination would influence the effective reproduction number by reducing the susceptible population.³⁹ Our basic reproduction number estimate was 1.17 (95% HPD, 1.06–1.30), toward the low end of prior

reports.^{16,17} We also consider two contextual factors: under-ascertainment, which, if roughly constant over the season, would mainly reduce observed counts rather than change growth, and dormitory/communal housing which may amplify local transmission. Our campus-wide estimate likely averages over these effects.

A limited number of recent studies have focused on sequencing and genomic epidemiology of seasonal influenza variants on college and university campuses since the COVID-19 pandemic. We performed surveillance on one of the largest universities in the United States during the 2022–2023 season and observed stark differences to national trends in clade diversity. We identified a local preference of A(H3N2) 3C.2a1b.2a.2a.3a.1 variants which did not assert its dominance of A(H3N2) on the national and international landscape until the following season. Through Bayesian inference and phylodynamics, we observed the impact of multiple domestic and international introductions fueling downstream transmission. Additionally, our use of epidemiologic modeling, epitope-based vaccine effectiveness estimates, and amino acid substitution profiling highlights the importance of antigenic drift in sustaining local influenza transmission dynamics. Taken together, we show the importance of genomic epidemiology in semi-closed, highly dense university settings and its potential for early insight into HA clade dominance on a national scale.

Despite its potential, more work is needed to promote campus-based influenza surveillance and integrate it with the state health agencies and CDC sentinel programs. Some universities, and even individual student health centers (e.g., Iowa State University's Thielen Student Health Center), participate in the National Respiratory and Enteric Virus Surveillance System⁴⁰ by reporting weekly aggregated virologic testing results. Weekly university influenza-like illness (ILI) counts can also be shared through ILINet.⁴¹ For sequencing and clade diversity, campuses could partner with their state public health laboratory to contribute a subset of positive specimens to the CDC's Virus Characterization program for genetic/antigenic characterization and antiviral susceptibility testing,⁴¹ while depositing their own consensus sequences to GenBank⁴² or GISAID.¹³ In parallel, participation in the CDC's Advanced Molecular Detection program including their Sequencing for Public Health Emergency Response, Epidemiology and Surveillance initiative⁴³ provides a community of practice for data sharing standards and best practices.

Limitations of the study

We note several limitations with this study including our focus on a single influenza season and one university student health clinic. Thus, we did not include results from asymptomatic individuals or those that sought care outside of the university's Tempe campus. We also did not include sample collection of individuals in the local Tempe, Phoenix, or Arizona-wide community for formal comparison. In addition, because we did not collect patient identifiers, we could not examine differences by sex or other demographic characteristics (e.g., age, race, and ethnicity).

We analyzed publicly available Arizona HA sequences from GISAID for the 2022–2023 season that lacked metadata indicating that they were from the university. Of 81 H3 sequences,

50 (62%) were 3C.2a1b.2a.2b and 10 (12%) were 3C.2a1b.2a.2a.3a.1, indicating that even state-level samples differed from our university H3 composition. Taken together, studies with expanded temporal and geographic scope are needed to strengthen generalizability for influenza surveillance.

We also acknowledge the potential of sampling bias to impact our Bayesian inference and epidemiological modeling. We attempted to reduce the impact of this by including representative H3 and N2 sequences from GISAID with consideration for diversity across genomic clades and subclades, collection date, and aggregated sampling location. In addition, we lacked metadata including campus residence halls and programs of study which limits our ability to study localized phylogenetics such as spread between residence halls. Including rich metadata in future studies (with ethics approval) would enable more refined analyses such as risk factors for intra-campus transmission, as seen in Holmes et al.⁴⁴ We also acknowledge that student vaccination data are not available for comparison to our epitope-driven VE estimates. Although, there is an on-campus health clinic that administers influenza vaccines, there are numerous pharmacies in the vicinity that also provide vaccination service.

We recognize that our use of the HANA assay developed by Zhou et al.⁴⁵ focuses on HA and NA segments and resulted in our high coverage for those segments (Figure S3), which is inconsistent with depth estimates for full-genome influenza assays.⁴⁶ Thus, because of our limitations in amplification and sequencing depth for internal gene segments, we were unable to perform additional analysis including evolutionary reassortment that might partially explain the multi-year difference in HA and NA TMRCA estimates.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Matthew Scotch (matthew.scotch@asu.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- We have deposited our assembled sequences into NCBI GenBank under accession numbers: PQ895943–PQ896387 (non-contiguous) for influenza A and PQ895532–PQ895537 for influenza B. In Table S6, we provide a list of the 503 sequences generated and used in this study.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

ACKNOWLEDGMENTS

The research reported in this publication was supported by the National Library of Medicine of the National Institutes of Health under award no. U01LM013129 to R.U.H., M.S., and A.V. The authors acknowledge the support from the ASU Health clinic staff and the Research Computing at Arizona State University for access and utilization of the Sol high-performance computing environment.⁴⁷ The authors also acknowledge all authors of GISAID sequences used in this manuscript.

AUTHOR CONTRIBUTIONS

Conceptualization, M.S.; methodology, M.S.; investigation, T.O.C.F. and J.M.W.; writing – original draft, M.S.; writing – review & editing, M.S., T.O.C.F., J.M.W., S.F., R.U.H., and A.V.; funding acquisition, M.S., R.U.H., and A.V.; resources, M.S., S.F., R.U.H., and A.V.; supervision, M.S. and T.O.C.F.

DECLARATION OF INTERESTS

The authors declare no conflicts of interests for the work described in this manuscript.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

We used an institutional license of ChatGPT 4o to generate the R code for visualization of Markov jump introductions and clade frequencies (Figures 2 and 3), lollipop graphs of amino acid substitutions (Figure 4), bar charts of median depth (Figure S3), SNP figures (Figures S4–S7), ANOVA calculations (Table S5), and monophyletic assessment of the MCC trees (Figures 2 and 3). We also used it to edit python code including for visualization of the MCC trees (Figures 2, 3, S13, and S14). After using this tool or service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **KEY RESOURCES TABLE**
- **EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS**
 - Ethics statement and biosafety
- **METHOD DETAILS**
 - Sample processing and molecular testing
 - Sequencing, assembly, variant calling, and similarity analysis
 - Bayesian phylogenetic and phylodynamic inference
 - Bayesian epidemiological modeling
 - Amino acid substitution profiles, vaccine protection estimates, and glycosylation sites
- **QUANTIFICATION AND STATISTICAL ANALYSIS**
 - Statistical methods

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.isci.2025.113941>.

Received: June 30, 2025

Revised: August 21, 2025

Accepted: October 30, 2025

Published: November 4, 2025

REFERENCES

1. Hill, V., Ruis, C., Bajaj, S., Pybus, O.G., and Kraemer, M.U.G. (2021). Progress and challenges in virus genomic epidemiology. *Trends Parasitol.* 37, 1038–1049. <https://doi.org/10.1016/j.pt.2021.08.007>.
2. Hill, V., Githinji, G., Vogels, C.B.F., Bento, A.I., Chaguza, C., Carrington, C.V.F., and Grubaugh, N.D. (2023). Toward a global virus genomic surveillance network. *Cell Host Microbe* 31, 861–873. <https://doi.org/10.1016/j.chom.2023.03.003>.
3. CDC (2024). Types of Influenza Viruses. <https://www.cdc.gov/flu/about/viruses-types.html>.
4. CDC (2024). Key Facts About Seasonal Flu Vaccine. <https://www.cdc.gov/flu/vaccines/keyfacts.html>.
5. Mostafa, A., Naguib, M.M., Nogales, A., Barre, R.S., Stewart, J.P., García-Sastre, A., and Martínez-Sobrido, L. (2024). Avian influenza A (H5N1) virus in dairy cattle: origin, evolution, and cross-species transmission. *mBio* 15, e0254224. <https://doi.org/10.1128/mbio.02542-24>.
6. Bouvier, N.M., and Palese, P. (2008). The biology of influenza viruses. *Vaccine* 26, D49–53. <https://doi.org/10.1016/j.vaccine.2008.07.039>.
7. Relich, R.F., and Loeffelholz, M.J. (2023). Taxonomic Changes for Human Viruses, 2020 to 2022. *J. Clin. Microbiol.* 61, e0033722. <https://doi.org/10.1128/jcm.00337-22>.
8. Maqsood, R., Smith, M.F., Holland, L.A., Sullins, R.A., Holland, S.C., Tan, M., Hernandez Barrera, G.M., Thomas, A.W., Islas, M., Kramer, J.L., et al. (2024). Influenza Virus Genomic Surveillance, Arizona, USA, 2023–2024. *Viruses* 16, 692. <https://doi.org/10.3390/v16050692>.
9. Chu, H.Y., Boeckh, M., Englund, J.A., Famulare, M., Lutz, B., Nickerson, D.A., Rieder, M., Starita, L.M., Adler, A., Brandstetter, E., et al. (2020). The Seattle Flu Study: a multiarm community-based prospective study protocol for assessing influenza prevalence, transmission and genomic epidemiology. *BMJ Open* 10, e037295. <https://doi.org/10.1136/bmjopen-2020-037295>.
10. Lewis, N.M., Delahoy, M.J., Sumner, K.M., Luring, A.S., Bendall, E.E., Mortenson, L., Edwards, E., Stamper, A., Flannery, B., and Martin, E.T. (2023). Risk factors for infection with influenza A(H3N2) virus on a US university campus, October–November 2021. *Influenza Other Respir. Viruses* 17, e13151. <https://doi.org/10.1111/irv.13151>.
11. ASU (2025). Facts and figures. <https://asu.edu/about/facts-and-figures>.
12. Hadfield, J., Megill, C., Bell, S.M., Huddleston, J., Potter, B., Callender, C., Sagulenko, P., Bedford, T., and Neher, R.A. (2018). Nextstrain: real-time tracking of pathogen evolution. *Bioinformatics* 34, 4121–4123. <https://doi.org/10.1093/bioinformatics/bty407>.
13. Shu, Y., and McCauley, J. (2017). GISAID: Global initiative on sharing all influenza data - from vision to reality. *Euro Surveill.* 22, 30494. <https://doi.org/10.2807/1560-7917.ES.2017.22.13.30494>.
14. CDC (2023). Weekly U.S. influenza surveillance report: key updates for week 20, ending May 20, 2023. <https://stacks.cdc.gov/view/cdc/129058>.
15. Ondo, G.N., Ushijima, Y., Abe, H., Mahmoudou, S., Bikangui, R., Nkoma, A.M., Mbadinga, M.J.V.M., More, A., Agbanrin, M.D., Pemba, C.M., et al. (2024). Genetic Diversity and Detection of Respiratory Viruses Excluding SARS-CoV-2 during the COVID-19 Pandemic in Gabon, 2020–2021. *Viruses* 16, 698. <https://doi.org/10.3390/v16050698>.
16. Biggerstaff, M., Cauchemez, S., Reed, C., Gambhir, M., and Finelli, L. (2014). Estimates of the reproduction number for seasonal, pandemic, and zoonotic influenza: a systematic review of the literature. *BMC Infect. Dis.* 14, 480. <https://doi.org/10.1186/1471-2334-14-480>.
17. Uyeki, T.M., Hui, D.S., Zambon, M., Wentworth, D.E., and Monto, A.S. (2022). Influenza. *Lancet* 400, 693–706. [https://doi.org/10.1016/S0140-6736\(22\)00982-5](https://doi.org/10.1016/S0140-6736(22)00982-5).
18. McCauley, J., Lewis, N., Daniels, R., Harvey, R., Galiano, M., Xiang, Z., Whittaker, L., Ermet, B., Cross, K., Rattigan, A., et al. (2023). Report prepared for the WHO annual consultation on the composition of influenza vaccines for the Southern Hemisphere. https://www.crick.ac.uk/sites/default/files/2022-10/Crick%20report%20Sep2022%20for%20SH2023_to%20post.pdf.
19. Gupta, R., and Brunak, S. (2002). Prediction of glycosylation across the human proteome and the correlation to protein function. *Pac. Symp. Bio-comput.* 7, 310–322.
20. European Centre for Disease Prevention and Control (2018). https://www.ecdc.europa.eu/sites/default/files/documents/AER_for_2016-influenza-seasonal.pdf.
21. CDC (2024). Weekly U.S. influenza surveillance report: key updates for week 20, ending May 18, 2024. <https://www.cdc.gov/flu/weekly/weeklyarchives2023-2024/week20.htm>.

22. CDC (2025). Weekly US Influenza Surveillance Report: key updates for week 18, ending May 3, 2025. <https://www.cdc.gov/fluview/surveillance/2025-week-18.html>.
23. Bedford, T., Greninger, A.L., Roychoudhury, P., Starita, L.M., Famulare, M., Huang, M.L., Nalla, A., Pepper, G., Reinhardt, A., Xie, H., et al. (2020). Cryptic transmission of SARS-CoV-2 in Washington state. *Science* 370, 571–575. <https://doi.org/10.1126/science.abc0523>.
24. Phyu, W.W., Saito, R., Kyaw, Y., Lin, N., Win, S.M.K., Win, N.C., Ja, L.D., Htwe, K.T.Z., Aung, T.Z., Tin, H.H., et al. (2022). Evolutionary Dynamics of Whole-Genome Influenza A/H3N2 Viruses Isolated in Myanmar from 2015 to 2019. *Viruses* 14, 2414. <https://doi.org/10.3390/v14112414>.
25. Wang, X., Walker, G., Kim, K.W., Stelzer-Braid, S., Scotch, M., and Rawlinson, W.D. (2024). The resurgence of influenza A/H3N2 virus in Australia after the relaxation of COVID-19 restrictions during the 2022 season. *J. Med. Virol.* 96, e29922. <https://doi.org/10.1002/jmv.29922>.
26. Westgeest, K.B., Russell, C.A., Lin, X., Spronken, M.I.J., Bestebroer, T.M., Bahl, J., van Beek, R., Skepner, E., Halpin, R.A., de Jong, J.C., et al. (2014). Genomewide analysis of reassortment and evolution of human influenza A(H3N2) viruses circulating between 1968 and 2011. *J. Virol.* 88, 2844–2857. <https://doi.org/10.1128/JVI.02163-13>.
27. Dorji, T., Dorji, K., and Gyeltshen, S. (2024). Evolution of Influenza A(H3N2) Viruses in Bhutan for Two Consecutive Years, 2022 and 2023. *Influenza Other Respir. Viruses* 18, e70028. <https://doi.org/10.1111/irv.70028>.
28. Westgeest, K.B., de Graaf, M., Fourment, M., Bestebroer, T.M., van Beek, R., Spronken, M.I.J., de Jong, J.C., Rimmelzwaan, G.F., Russell, C.A., Osterhaus, A.D.M.E., et al. (2012). Genetic evolution of the neuraminidase of influenza A (H3N2) viruses from 1968 to 2009 and its correspondence to haemagglutinin evolution. *J. Gen. Virol.* 93, 1996–2007. <https://doi.org/10.1099/vir.0.043059-0>.
29. Anoma, S., Bhattachakosol, P., and Kowitdamrong, E. (2024). Characteristics and evolution of hemagglutinin and neuraminidase genes of Influenza A(H3N2) viruses in Thailand during 2015 to 2018. *PeerJ* 12, e17523. <https://doi.org/10.7717/peerj.17523>.
30. Ho, S.Y.W., Lanfear, R., Bromham, L., Phillips, M.J., Soubrier, J., Rodrigo, A.G., and Cooper, A. (2011). Time-dependent rates of molecular evolution. *Mol. Ecol.* 20, 3087–3101. <https://doi.org/10.1111/j.1365-294X.2011.05178.x>.
31. Rambaut, A., Pybus, O.G., Nelson, M.I., Viboud, C., Taubenberger, J.K., and Holmes, E.C. (2008). The genomic and epidemiological dynamics of human influenza A virus. *Nature* 453, 615–619. <https://doi.org/10.1038/nature06945>.
32. Holmes, E.C., Ghedin, E., Miller, N., Taylor, J., Bao, Y., St George, K., Grenfell, B.T., Salzberg, S.L., Fraser, C.M., Lipman, D.J., and Taubenberger, J.K. (2005). Whole-genome analysis of human influenza A virus reveals multiple persistent lineages and reassortment among recent H3N2 viruses. *PLoS Biol.* 3, e300. <https://doi.org/10.1371/journal.pbio.0030300>.
33. Fratty, I.S., Jurkovic, M., Zuckerman, N., Nemet, I., Atari, N., Kliker, L., Gur-Arie, L., Rosenberg, A., Glatman-Freedman, A., Lustig, Y., and Mandelboim, M. (2023). Influenza vaccine compatibility among hospitalized patients during and after the COVID-19 pandemic. *Front. Microbiol.* 14, 1296179. <https://doi.org/10.3389/fmicb.2023.1296179>.
34. Galli, C., Mazzola, G., Arosio, M., Pellegrinelli, L., Boldrini, A., Guarneri, D., Lombarda, E., Farina, C., Cereda, D., and Pariani, E. (2024). Real-time investigation of an influenza A(H3N2) virus outbreak in a refugee community, November 2022. *Public Health* 230, 157–162. <https://doi.org/10.1016/j.puhe.2024.02.027>.
35. Brcko, I.C., de Souza, V.C., Ribeiro, G., Lima, A.R.J., Martins, A.J., Barros, C.R.D.S., de Carvalho, E., Pereira, J.S., de Lima, L.P.O., Viala, V.L., et al. (2025). Comprehensive molecular epidemiology of influenza viruses in Brazil: insights from a nationwide analysis. *Virus Evol.* 11, veae102. <https://doi.org/10.1093/ve/veae102>.
36. Korsun, N., Trifonova, I., Pavlova, D., Uzunova, Y., Ivanov, I., Ivanov, D., Velikov, P., Voleva, S., Tcherveniakova, T., and Christova, I. (2025). Etiological Spectrum of Acute Respiratory Infections in Bulgaria During the 2023–2024 Season and Genetic Diversity of Circulating Influenza Viruses. *Viruses* 17, 270. <https://doi.org/10.3390/v17020270>.
37. CDC (2025). CDC Seasonal Flu Vaccine Effectiveness Studies. <https://www.cdc.gov/flu-vaccines-work/php/effectiveness-studies/index.html>.
38. Maurel, M., Pozo, F., Pérez-Gimeno, G., Buda, S., Sève, N., Oroszi, B., Hooiveld, M., Gomez, V., Domegan, L., Martínez-Baz, I., et al. (2024). Influenza vaccine effectiveness in Europe: Results from the 2022–2023 VEBIS (Vaccine Effectiveness, Burden and Impact Studies) primary care multi-centre study. *Influenza Other Respir. Viruses* 18, e13243. <https://doi.org/10.1111/irv.13243>.
39. Delamater, P.L., Street, E.J., Leslie, T.F., Yang, Y.T., and Jacobsen, K.H. (2019). Complexity of the Basic Reproduction Number ($R(0)$). *Emerg. Infect. Dis.* 25, 1–4. <https://doi.org/10.3201/eid2501.171901>.
40. CDC (2025). National Respiratory and Enteric Virus Surveillance System (NREVSS): Participating Labs. <https://www.cdc.gov/nrevss/php/participating-labs/index.html>.
41. CDC (2025). U.S. Influenza Surveillance: Purpose and Methods. <https://www.cdc.gov/fluview/overview/index.html>.
42. Sayers, E.W., Cavanaugh, M., Frisse, L., Pruitt, K.D., Schneider, V.A., Underwood, B.A., Yankie, L., and Karsch-Mizrachi, I. (2025). GenBank 2025 update. *Nucleic Acids Res.* 53, D56–D61. <https://doi.org/10.1093/nar/gkaf1114>.
43. CDC (2024). 3-Year Anniversary of National Viral Genomics Consortium. <https://www.cdc.gov/advanced-molecular-detection/php/success-stories/spheres.html>.
44. Holmes, E.C., Ghedin, E., Halpin, R.A., Stockwell, T.B., Zhang, X.Q., Fleming, R., Davey, R., Benson, C.A., Mehta, S., Taplitz, R., et al. (2011). Extensive geographical mixing of 2009 human H1N1 influenza A virus in a single university community. *J. Virol.* 85, 6923–6929. <https://doi.org/10.1128/JVI.00438-11>.
45. Zhou, B., Deng, Y.M., Barnes, J.R., Sessions, O.M., Chou, T.W., Wilson, M., Stark, T.J., Volk, M., Spirason, N., Halpin, R.A., et al. (2017). Multiplex Reverse Transcription-PCR for Simultaneous Surveillance of Influenza A and B Viruses. *J. Clin. Microbiol.* 55, 3492–3501. <https://doi.org/10.1128/JCM.00957-17>.
46. Zhao, J., Liu, J., Vemula, S.V., Lin, C., Tan, J., Ragupathy, V., Wang, X., Mbondji-Wonje, C., Ye, Z., Landry, M.L., and Hewlett, I. (2016). Sensitive Detection and Simultaneous Discrimination of Influenza A and B Viruses in Nasopharyngeal Swabs in a Single Assay Using Next-Generation Sequencing-Based Diagnostics. *PLoS One* 11, e0163175. <https://doi.org/10.1371/journal.pone.0163175>.
47. Jennewein, D.M., Lee, J., Kurtz, C., Dizon, W., Shaeffer, I., Chapman, A., Chiquete, A., Burks, J., Carlson, A., Mason, N., et al. (2023). The Sol Super-computer at Arizona State University. Held in Portland, OR USA (ACM), pp. 296–301.
48. WHO (2025). WHO information for the molecular detection of influenza viruses. <https://www.who.int/teams/global-influenza-programme/laboratory-network/quality-assurance/eqa-project/information-for-molecular-diagnosis-of-influenza-virus>.
49. Suchard, M.A., Lemey, P., Baele, G., Ayres, D.L., Drummond, A.J., and Rambaut, A. (2018). Bayesian phylogenetic and phylodynamic data integration using BEAST 1.10. *Virus Evol.* 4, vey016. <https://doi.org/10.1093/ve/vey016>.
50. Volz, E.M., and Siveroni, I. (2018). Bayesian phylodynamic inference with complex models. *PLoS Comput. Biol.* 14, e1006546. <https://doi.org/10.1371/journal.pcbi.1006546>.
51. Bouckaert, R., Heled, J., Kühnert, D., Vaughan, T., Wu, C.H., Xie, D., Suchard, M.A., Rambaut, A., and Drummond, A.J. (2014). BEAST 2: a software platform for Bayesian evolutionary analysis. *PLoS Comput. Biol.* 10, e1003537. <https://doi.org/10.1371/journal.pcbi.1003537>.
52. Kearse, M., Moir, R., Wilson, A., Stones-Havas, S., Cheung, M., Sturrock, S., Buxton, S., Cooper, A., Markowitz, S., Duran, C., et al. (2012). Geneious Basic: an integrated and extendable desktop software platform for the

- p>organization and analysis of sequence data.
- Bioinformatics*
- 28, 1647–1649.
- <https://doi.org/10.1093/bioinformatics/bts199>
- .
53. Rambaut, A., Drummond, A.J., Xie, D., Baele, G., and Suchard, M.A. (2018). Posterior Summarization in Bayesian Phylogenetics Using Tracer 1.7. *Syst. Biol.* 67, 901–904. <https://doi.org/10.1093/sysbio/syy032>.
 54. Paradis, E., and Schliep, K. (2019). ape 5.0: an environment for modern phylogenetics and evolutionary analyses in R. *Bioinformatics* 35, 526–528. <https://doi.org/10.1093/bioinformatics/bty633>.
 55. Schliep, K.P. (2011). phangorn: phylogenetic analysis in R. *Bioinformatics* 27, 592–593. <https://doi.org/10.1093/bioinformatics/bty706>.
 56. Wang, L.G., Lam, T.T.Y., Xu, S., Dai, Z., Zhou, L., Feng, T., Guo, P., Dunn, C.W., Jones, B.R., Bradley, T., et al. (2020). Treeio: An R Package for Phylogenetic Tree Input and Output with Richly Annotated and Associated Data. *Mol. Biol. Evol.* 37, 599–603. <https://doi.org/10.1093/molbev/msz240>.
 57. Wickham, H., François, R., Henry, L., Müller, K., and Vaughan, D. (2025). dplyr: A Grammar of Data Manipulation. <https://dplyr.tidyverse.org/>.
 58. Wickham, H. (2016). ggplot2 : Elegant Graphics for Data Analysis. Use R!, 2nd ed. (Springer International Publishing: Imprint: Springer).
 59. Dudas, G. (2023). baltic. <https://github.com/evogytis/baltic/>.
 60. Shepard, S.S., Meno, S., Bahl, J., Wilson, M.M., Barnes, J., and Neuhaus, E. (2016). Viral deep sequencing needs an adaptive approach: IRMA, the iterative refinement meta-assembler. *BMC Genom.* 17, 708. <https://doi.org/10.1186/s12864-016-3030-6>.
 61. Katoh, K., Misawa, K., Kuma, K.i., and Miyata, T. (2002). MAFFT: a novel method for rapid multiple sequence alignment based on fast Fourier transform. *Nucleic Acids Res.* 30, 3059–3066. <https://doi.org/10.1093/nar/gkf436>.
 62. Rambaut, A., Lam, T.T., Max Carvalho, L., and Pybus, O.G. (2016). Exploring the temporal structure of heterochronous sequences using TempEst (formerly Path-O-Gen). *Virus Evol.* 2, vew007. <https://doi.org/10.1093/ve/vew007>.
 63. Ou, J., and Zhu, L.J. (2019). trackViewer: a Bioconductor package for interactive and integrative visualization of multi-omics data. *Nat. Methods* 16, 453–454. <https://doi.org/10.1038/s41592-019-0430-y>.
 64. Lawrence, M., Huber, W., Pagès, H., Aboyoun, P., Carlson, M., Gentleman, R., Morgan, M.T., and Carey, V.J. (2013). Software for computing and annotating genomic ranges. *PLoS Comput. Biol.* 9, e1003118. <https://doi.org/10.1371/journal.pcbi.1003118>.
 65. Muhire, B.M., Varsani, A., and Martin, D.P. (2014). SDT: a virus classification tool based on pairwise sequence alignment and identity calculation. *PLoS One* 9, e108277. <https://doi.org/10.1371/journal.pone.0108277>.
 66. Bonomo, M.E., and Deem, M.W. (2018). Predicting Influenza H3N2 Vaccine Efficacy From Evolution of the Dominant Epitope. *Clin. Infect. Dis.* 67, 1129–1131. <https://doi.org/10.1093/cid/ciy323>.
 67. Maurer-Stroh, S., Lee, R.T.C., Limvipuvadh, V., Ma, J., Sirota, F.L., Gunalan, V., Paing, S.S.T., Doungpan, N., Xiang, J., and Eisenhaber, F. (2025). FluServer: Real-time surveillance of influenza mutations. <http://fluserver.bii.a-star.edu.sg>.
 68. Aksamentov, I., Roemer, C., Hodcroft, E., and Neher, R. (2021). Nextclade: clade assignment, mutation calling and quality control for viral genomes. *J. Open Source Softw.* 6, 3773. <https://doi.org/10.21105/joss.03773>.
 69. Neher, R.A., Aksamentov, I., and Roemer, C. (2025). Nextclade Web. <https://clades.nextstrain.org/>.
 70. Bao, Y., Bolotov, P., Dernovoy, D., Kiryutin, B., and Tatusova, T. (2007). FLAN: a web server for influenza virus genome annotation. *Nucleic Acids Res.* 35, W280–W284. <https://doi.org/10.1093/nar/gkm354>.
 71. De Coster, W., and Rademakers, R. (2023). NanoPack2: population-scale evaluation of long-read sequencing data. *Bioinformatics* 39, btad311. <https://doi.org/10.1093/bioinformatics/btad311>.
 72. Galaxy Community (2024). The Galaxy platform for accessible, reproducible, and collaborative data analyses: 2024 update. *Nucleic Acids Res.* 52, W83–W94. <https://doi.org/10.1093/nar/gkae410>.
 73. Andrews, S. (2024). FastQC. <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>.
 74. Wick, R.R. (2018). Porechop. <https://github.com/rwick/Porechop>.
 75. Chen, S., Zhou, Y., Chen, Y., and Gu, J. (2018). fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics* 34, i884–i890. <https://doi.org/10.1093/bioinformatics/bty560>.
 76. Li, H. (2018). Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics* 34, 3094–3100. <https://doi.org/10.1093/bioinformatics/bty191>.
 77. Oxford Nanopore Technologies (2024). Medaka. <https://github.com/nanoporetech/medaka>.
 78. Danecek, P., Bonfield, J.K., Liddle, J., Marshall, J., Ohan, V., Pollard, M.O., Whitwham, A., Keane, T., McCarthy, S.A., Davies, R.M., and Li, H. (2021). Twelve years of SAMtools and BCFtools. *GigaScience* 10, giab008. <https://doi.org/10.1093/gigascience/giab008>.
 79. Faleye, T.O.C., Adams, D., Adhikari, S., Sandrolini, H., Halden, R.U., Varsani, A., and Scotch, M. (2021). Use of hemagglutinin and neuraminidase amplicon-based high-throughput sequencing with variant analysis to detect co-infection and resolve identical consensus sequences of seasonal influenza in a university setting. *BMC Infect. Dis.* 21, 810. <https://doi.org/10.1186/s12879-021-06526-5>.
 80. Hill, V., and Baele, G. (2019). Bayesian Estimation of Past Population Dynamics in BEAST 1.10 Using the Skygrid Coalescent Model. *Mol. Biol. Evol.* 36, 2620–2628. <https://doi.org/10.1093/molbev/msz172>.
 81. Baele, G., Lemey, P., Bedford, T., Rambaut, A., Suchard, M.A., and Alekseyenko, A.V. (2012). Improving the accuracy of demographic and molecular clock model comparison while accommodating phylogenetic uncertainty. *Mol. Biol. Evol.* 29, 2157–2167. <https://doi.org/10.1093/molbev/mss084>.
 82. Lemey, P., Rambaut, A., Drummond, A.J., and Suchard, M.A. (2009). Bayesian phylogeography finds its roots. *PLoS Comput. Biol.* 5, e1000520. <https://doi.org/10.1371/journal.pcbi.1000520>.
 83. Minin, V.N., and Suchard, M.A. (2008). Counting labeled transitions in continuous-time Markov models of evolution. *J. Math. Biol.* 56, 391–412. <https://doi.org/10.1007/s00285-007-0120-8>.
 84. HHS (2025). HHS Regional Offices. <https://www.hhs.gov/about/agencies/iea/regional-offices/index.html>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological samples		
nasopharyngeal (NP) or mid-turbinate swabs	Arizona State University student health services (see <i>ethics statement</i>)	
Critical commercial assays		
QIAamp Viral RNA kit	Qiagen, Inc., Venlo, Netherlands	
SuperScript IV first strand cDNA synthesis kit	Thermo Fisher, Waltham, MA, USA	
SsoAdvanced Universal Probes Supermix	Bio-Rad, Hercules, CA, USA	
Phusion-plus master mix	Thermo Fisher, Waltham, MA, USA	
MinION SQK-LSK110 and EXP-NBD104 library kits	Oxford Nanopore Technologies, Oxford, UK	
Deposited data		
GenBank	This study. See Table S6 .	PQ895943–PQ896387 (non-contiguous) for influenza A and PQ895532–PQ895537 for influenza B.
Oligonucleotides		
Influenza A and B, HA and NA primers and probes	Zhou et al. ⁴⁵ and WHO ⁴⁸	
Software and algorithms		
BEAST v1.10.4, BEAUti 1.10.4, LogCombiner 1.10.4, TreeAnnotator 1.10.4	Suchard et al. ⁴⁹	
PhyDyn, BEAST2	Volz and Siveroni ⁵⁰ ; Bouckaert et al. ⁵¹	
Geneious Prime v2024.0.7	Dotmatics, Boston, MA USA and Kearse et al. ⁵²	
Tracer v1.7	Rambaut et al. ⁵³	
ape, R and RStudio 2024.12.1	Paradis and Schliep ⁵⁴	
phangorn, R and RStudio 2024.12.1	Schliep ⁵⁵	
treeio, R and RStudio 2024.12.1	Wang et al. ⁵⁶	
dplyr, R and RStudio 2024.12.1	Wickham et al. ⁵⁷	
ggplot2, R and RStudio 2024.12.1	Dudas ⁵⁸	
baltic, Python3	Dudas ⁵⁹	
IRMA v1.1.4, CDC	Shepard et al. ⁶⁰	
MAFFT v7.490	Geneious Prime v2024.0.7 Kearse et al. ⁵² and MAFFT Katoh et al. ⁶¹	
TempEST v1.5.3	Rambaut et al. ⁶²	
trackViewer, R and RStudio 2024.12.1	Ou and Zhu ⁶³	
GenomicRanges, R and RStudio 2024.12.1	Lawrence et al. ⁶⁴	
Sequence Demarcation Tool (SDT) v1.3	Muhire ⁶⁵	
pEpitope v2.2 in MATLAB	Bonomo and Deem ⁶⁶	
FluServer	Maurer-Stroh et al. ⁶⁷	
NetNGlyc-1.0	Gupta and Brunak ¹⁹	
Nextclade	Aksamentov et al. ⁶⁸ ; Neher ⁶⁹	
Nextstrain	Hadfield et al. ¹²	
FLAN, NCBI	Bao et al. ⁷⁰	
NanoPlot v1.44.1+galaxy0	De Coster and Rademakers ⁷¹ 2023 ⁷²	
FastQC v0.74+galaxy1	Galaxy Community ⁷² and Andrews ⁷³	
Porechop v0.2.4	Wick et al. ⁷⁴	

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
fastplong v0.2.2	Chen et al. ⁷⁵	
Minimap2 v2.29	Li ⁷⁶	
Medaka v1.12.0	⁷⁷	
snEff v5.2+galaxy0	Galaxy Community ⁷² ; Danecek et al. ⁷⁸	
bcftools v1.15.1+galaxy4	Galaxy Community ⁷² ; Danecek et al. ⁷⁸	

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Ethics statement and biosafety

Arizona State University (ASU) Institutional Review Board (IRB) approved *Surveillance of Influenza on ASU Campus* (STUDY00008985) on October 11, 2018. The project was later determined to be non-human subjects through a revised submission focusing on samples only (STUDY00019217) on February 5, 2024. Nasopharyngeal (NP) or mid-turbinate swabs were collected as part of routine clinical care from patients presenting to the student health clinic with influenza-like illness (ILI). In addition, only influenza test result and collection date were recorded for the study and thus no personal health information was utilized. The ASU Institutional Biosafety Committee (IBC) approved the protocol (SPROTO202200000031) on 18 July 2022 at a BSL-2+ containment level.

METHOD DETAILS

Sample processing and molecular testing

As part of an on-going surveillance study, we analyzed nasopharyngeal (NP) or mid-turbinate swabs from student health services on the Tempe campus of Arizona State University (ASU) during the 2022–2023 North American influenza season. The clinic staff tested students via a lateral flow immunoassay as part of routine medical care who presented with influenza-like illness (ILI). They labeled the collection tubes with the result (either influenza A, B, A/B, or *negative*) and provided the date of sample collection.

At our laboratory on campus, we froze samples at -80°C or immediately subjected 140 μL to RNA extraction with the QIAamp Viral RNA kit according to the manufacturer's specifications (Qiagen, Inc., Venlo, Netherlands). We performed cDNA synthesis on the extracted RNA using the universal influenza A/B UniFlu primer (5'-IAGCARAAGC-3') described by Zhao et al.⁴⁶ and the SuperScript IV first strand cDNA synthesis kit (Thermo Fisher, Waltham, MA, USA) following the manufacturer's instructions. We stored the cDNA at -20°C and used it for downstream assays.

We performed PCR to confirm detection of influenza A and B nucleic acid using the SsoAdvanced Universal Probes Supermix (Bio-Rad, Hercules, CA, USA) following the manufacturer's instructions. For amplification, we used a QuantStudio 5 thermocycler (Thermo Fisher, Waltham, MA, USA) with reaction settings that include an initial denaturation at 95°C for 3 min, followed by 40 cycles of 95°C for 15 s and 60°C for 30 s during which time we recorded fluorescence data.

For confirmed positive samples, we used the multiplex assay by Zhou et al.,⁴⁵ and as previously described by,⁷⁹ for amplification of the hemagglutinin (HA) and neuraminidase (NA) coding segments on a Bio-Rad C1000 thermal cycler (Bio-Rad, Hercules, CA, USA). We used the cDNA along with the forward and reverse primers from Table S7 and the Phusion-plus master mix (Thermo Fisher, Waltham, MA, USA) with the following thermal cycling conditions; 94°C for 2 min, 5 cycles of 94°C for 30 s, 45°C for 30 s, and 68°C for 3 min; 35 cycles of 94°C for 30 s, 57°C for 30 s, and 68°C for 3 min; followed by a final elongation step of 68°C for 10 min and 12°C until stopped. Finally, we purified the amplicons via magnetic beads, and prepared MinION long-read sequencing libraries (MinION SQK-LSK110 and EXP-NBD104, Oxford Nanopore Technologies, Oxford, UK) following the manufacturer's instructions.

Sequencing, assembly, variant calling, and similarity analysis

We sequenced our long-read libraries on a MinION Mk1b with a flongle R9.4.1 flow cell (Oxford Nanopore, Oxford, UK) and utilized Guppy v6.2.7 (as part of MinKNOW v22.08.4) for basecalling. We used FastQC v0.74⁷³ and NanoPlot v1.44.1⁷¹ on galaxy⁷² for quality assurance. We trimmed adapters from the raw reads using porechop v0.2.4⁷⁴ and filtered out reads with <500 nucleotides and phred scores <15 using fastplong v0.2.2.⁷⁵ We re-checked for QC using FastQC and NanoPlot and assembled them via the Centers for Disease Control and Prevention's (CDC) IRMA v1.1.4 pipeline using the *FLU-minion* module⁶⁰ and Minimap2 v2.29 for assembly.⁷⁶ For our reference set, we used the recommended 2022–2023 Northern Hemisphere vaccine sequences (Table S8). For the remaining steps in our pipeline, we used Medaka v1.12.0 for polishing and variant analysis,⁷⁷ SnpEff v5.2 (on galaxy) for annotation and functional effect prediction,⁷⁸ and bcftools v 1.15.1 (on galaxy) or querying relevant columns in the variant files.⁷⁸ Finally, we used the NCBI's Influenza Virus Sequence Annotation Tool⁷⁰ for subtype confirmation and identification of any remaining errors in the consensus sequence.

We uploaded our consensus FASTA sequences to Nextclade^{68,69} and removed, from our datasets, any sequences receiving an overall quality status of *bad*. For initial comparison, we used the Sequence Demarcation Tool (SDT) v1.3⁶⁵ for sequencing similarity analysis.

Bayesian phylogenetic and phylodynamic inference

We combined our new sequences with 2022–2023 sequences from GISAID to create segment and subtype-specific datasets. We performed multiple sequence alignment in Geneious Prime v.2025.1.2⁵² (Dotmatics, Boston, MA USA) using MAFFT v7.490⁶¹ with default parameters and visually inspected the resulting alignment to identify and remove any misclassified sequences or alignment errors.

We used TempEST v1.5.3⁵³ to assess the temporal signal and identified a strong clock-like relationship between sampling time and root-to-tip divergence in both the H3 ($R^2 > 0.92$) and N2 ($R^2 > 0.87$) phylogenies under the best-fitting root; supporting the use of a strict molecular clock model.

We used BEAUti v1.10.4⁴⁹ to specify different population growth priors including a constant growth and a non-parametric Bayesian skygrid coalescent model.⁸⁰ We evaluated their fit to the data by calculating the log-marginal likelihood values via stepping-stone and path-sampling analysis⁸¹ which suggested the use of the nonparametric skygrid under a strict clock for both H3 and N2 datasets (Table S9). We ran separate Markov chain Monte Carlo (MCMC) simulations for 5×10^7 steps and sampled every 5×10^3 steps. We combined posterior data via LogCombiner v1.10.4⁴⁹ and checked for convergence via Tracer v1.7.1⁵³ and that all effective sampling size (ESS) values were ≥ 200 . We generated a maximum clade credibility (MCC) tree via TreeAnnotator v1.10.4⁴⁹ from the combined set of posterior trees.

To understand the spatiotemporal and evolutionary dynamics of influenza H3 clades and N2 subclades circulating at the university, including the number and timing of migration events, we selected 1,000 trees from the combined posterior distribution and used them as empirical input for discrete trait ancestral state reconstruction in BEAST v1.10.4.⁴⁹ We annotated each sequence with a discrete location attribute corresponding to its sampling region (e.g., North America, Europe, Asia, university). We applied a discrete trait model with Bayesian stochastic search variable selection (BSSVS) to estimate asymmetric transition rates between locations while incorporating model uncertainty.⁸² To infer the geographic sources and timing of introductions into the university, we recorded Markov jumps from all regions to the university.⁸³ We ran the analysis for 2×10^6 steps, sampling every 2×10^2 steps. After discarding 10% burn-in, we summarized the posterior tree set using TreeAnnotator v1.10.4⁴⁹ to generate a maximum clade credibility (MCC) tree. From this tree, we used custom R scripts including *ape*,⁵⁴ *phangorn*,⁵⁵ *treeio*,⁵⁶ *dplyr*,⁵⁷ *trackViewer*,⁶³ *GenomicRanges*,⁶⁴ and *ggplot2*⁵⁸ packages to extract the number, timing, and geographic origin of introduction events, along with the number of descendant tips per introduction. We defined an introduction event as a branch in the MCC where a Markov jump to the university was inferred (i.e., toASU_median = 1) and the parent state was not the university. We estimated the timing of this event based on the midpoint of the related branch and summarized the size of the introduction by counting the number of descendant tips.

Bayesian epidemiological modeling

To understand the population dynamics of seasonal influenza on a large university campus, we leveraged the *PhyDyn* package as part of BEAST2.^{50,51} This approach allows for the simultaneous estimation of epidemiological-relevant parameters (such as R_0) and coalescent-based phylogenies.⁵⁰ The epidemic models in *PhyDyn* are implemented through ordinary differential equations (ODEs) to estimate the rates of new infections and transitions between states. We created an additional dataset of sequences from the 2022–2023 season via stratified sampling to consider representative coverage with both temporal and geographic diversity. First, we grouped sequences into weekly intervals and randomly selected up to ten sequences per week. Second, we proportionally sampled based on the availability of sequences from each of the ten Health and Human Services (HHS) regions.⁸⁴ To maximize diversity, we combined sequences sampled in both strategies and removed duplicates. In Geneious Prime v2024.0.7⁵² (Dotmatics, Boston, MA USA), we removed records that reflected incomplete coding regions (e.g., lacking a start and/or stop codon) and performed multiple sequence alignment via the MAFFT plugin v7.490⁶¹ with default settings.

For our H3 dataset, we utilized a two-deme SIR model adapted from Volz et al.⁵⁰ including our campus population and a global A(H3N2) community (see Methods S1).

Amino acid substitution profiles, vaccine protection estimates, and glycosylation sites

In Geneious Prime v2024.0.7⁵² (Dotmatics, Boston, MA USA), we translated our HA and NA gene segments into protein sequences and checked for variants and phenotypic importance via FluServer⁶⁷ by specifying the recommended vaccine strains for each subtype (Table S8). For each query, we examined the interest level (0–3) of the amino acid substitution, any previously reported effects or multiple position effects, and known structural interactions. For neuraminidase, we additionally considered whether there were substitutions with known antiviral resistance.

As the effectiveness of influenza vaccines (VE) changes from season-to-season, we analyzed the protection of the vaccine among our translated H3 hemagglutinin sequences via pEpitope v2.2⁶⁶ in MATLAB. Finally, we used the NetNGlyc-1.0 server¹⁹ to predict N-glycosylation sites which considers N-X-S/T motifs where X can be any amino acid except proline.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical methods

We describe all statistical analyses and quantitative measurement in the [results](#) section and the [supplemental information](#). We used BEAST v1.10.4 and BEAST v2.7.7 for Bayesian phylogenetic and phylodynamic inference and used Tracer v1.7.1 to verify that all

parameters had effective sample sizes ($ESS \geq 200$). We compared log marginal likelihoods via stepping-stone and path-sampling metrics, with results in the [supplemental information \(Table S9\)](#). For posterior estimates such as the most recent common ancestor (TMRCA), we report medians and highest posterior density (HPD) intervals as measures of central tendency and precision.

We used the PhyDyn package in BEAST v2.7.7 to estimate epidemiological parameters such as the basic reproduction number (R_0) via a two-deme SIR model adapted from Volz et al.⁵⁰ Here, we used ordinary differential equations to describe changes in susceptible, infected, and recovered populations, with initial values and model equations provided in the [supplemental information \(Methods S1\)](#).

We used R and RStudio 2024.12.1 to evaluate differences in predicted vaccine effectiveness through a one-way ANOVA using the `aov()` function ([Table S5](#)) and a statistical significance of $p < 0.05$. We used NetNGlyc v1.0 to predict N-glycosylation sites and using mean potential cutoffs of 0.5, 0.75, and 0.90, as well as jury agreement to categorize results ([Table 3](#)).

Unless otherwise indicated, n refers to the number of independent influenza virus sequences analyzed (e.g., $n = 91$ for H3 and $n = 76$ for N2).